MILITARY SPECIFICATION

FILTER, AIR, CHEMICAL-BIOLOGICAL: M46

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of filter for use in the emergency back-up protective filtering system of the toxicological micro-climate controlled protective outfit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced-
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-B-46506 - Box, Ammunition Packing Wood, Wirebound.

US GOVERNMENT PRINTING OFFICE 1981-703-023 5196 FSC 4240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Reserach and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. *

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads for 40″ by 48″ Pallets, Skids, Runners or Pallet-Type Base.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D5-3-951 - Filter, Air, Chemical-Biological: M46.
C136-301-73 - Inhalation Resistance Test Fixture.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D412 - Rubber Properties in Tension.
D2240 - Rubber Property - Durometer Hardness.

(Application for copies of the above publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials and components.

3.1. 1 Materials. All materials cited on Drawing D5-3-951 and it subsidiary drawings shall conform to the specifications listed thereon.

3.1.2 Components. All components of the filter shall conform to the specifications listed on Drawing D5-3-951 and its subsidiary drawings.

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3.2 Manufacture and assembly. The filter shall be manufactured and assembled to conform to Drawing D5-3-951.

3.3 Leakage. The assembled filter without filter elements installed but including the quick disconnect junction shall not show evidence of leakage when pressurized with air to 50.0 ± 5 inches of water and submerged in water and tested as specified in 4.4.4.1.

3.4 Inhalation resistance. Inhalation resistance of the assembled filter with the filter element set (see 4.1.2) shall be no less than 40 and no more than 65 millimeters of water at a continuous inhalation of 85 liters per minute when tested as specified in 4.4.4.2.

3.5 Workmanship. The assembled filter without the filter element set installed shall be free from contamination and imperfections such as grease, oil, dirt, foreign matter, damage or distortion of metal parts, malfunctioning hardware and in addition the rubber products shall be free of defects specified in MIL-STD-407.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of filter shall be produced in accordance with this specification for examination and test (see 4.3).

3.7 Physical properties of molded parts. The molded butyl rubber front and back housings shall meet the durometer hardness (Shore A) and tensile strength requirements of Grade RS 615 of MIL-STD-417 when tested as specified in 4.4.4.3*

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government's responsibility. The Government will be responsible for furnishing the filter element set, chemical-biological mask, M13 and M13A1 used during testing.

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting office that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspecitons. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction.

4.3.1 Sample. The preproduction of 50 filters shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Preproduction sample of filter shall be inspected for compliance with the drawings and all the characteristics included in the classification of defects (see 4.4.3.4). The inhalation resistance test (4.4.4.2) shall be performed prior to any pressure leakage testing (4.4.4.1).

4.3.2.2 For destructive tests. Five back housings and five front housings shall be taken randomly from the preproduction samples before assembly arid tested as specified in 4.4.4.3. The acceptance number shall be zero.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the inspection procedure specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A regular production lot shall consist of the filter produced by one manufacturer at one plant from the same materials and under essentially the same manufacturing conditions. Any change in equipment or procedure used in regular production from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Five back housings and five front housings shall be taken randomly from the lot before assembly and tested as specified in 4.4.4.3.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample filter shall be examined and tested in accordance with the classification of defects and with MIL-STD-105. The inhalation resistance test (4.4.4.2) shall be performed prior to any pressure leakage testing (4.4.4.1).

4.4.3.2 For destructive tests. The back and front housings shall be tested for physical properties as specified in 4.4.4.3 to meet the requirement of 3.7. The acceptance number is zero.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects*

4.4.3.4 Classification of defects.

(a) Filter, air, chemical-biological: M46 (Drawing D5-3-951).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Major defects listed in MIL-STD-407	
103	Marking incorrect	
104	Workmanship not in MIL-STD-407 (3.5)	
105	Inhalation resistance	4.4.4.2
Minor:	AQL 4.0 percent defective	
201	Minor defects listed in MIL-STD-407	

(b) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Container incorrect
102	Package incorrectly assembled
103	Container incorrectly closed
104	Marking missing, incorrect or illegible

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4.4.4 Tests.

4.4.4.1 Leakage. The assembled filter with emergency breathing adapter (see 6.3) attached shall be pressurized with air to 50.0 ± 5 inches of water and submerged just below the surface of the water (at room temperature) in each of two orientations (one with front facing upwards and one with front facing downwards) for at least 30 seconds each. Bubbles rising from this assembly after 15 seconds of submersion for either orientation shall be cause of rejection The filter shall be thoroughly dried at the completion of the test.

4.4.4.2 Inhalation resistance. Inhalation resistance of the assembled filter with filter element set shall be measured using the Inhalation Resistance Test Fixture conforming to Drawing C136-301-73.

4.4.4.3 Physical properties of molded parts. The front and back housings shall be tested for tensile strength in accordance with ASTM D412 die D and durometer hardness in accordance with ASTM D 2240.

5. PACKAGING

5.1 Preservation (level A and B). Prior to unit packing (5.2.1 and 5.2.2), loosen the two screws on clamp assembly to alleviate pressure on rubber housing.

5.1.1 Cleaning. The filter shall be free of dirt and other foreign matter with unit packed.

5.2 Unit packing. Level shall be as specified (see 6.2).

5.2.1 Level A. The filter shall be unit packed in accordance with Drawing D5-3-1008.

5.2.2 Level B. The filter shall be unit packed, one each, in a fiberboard box conforming to style FOL, grade W5c, PPP-B-636, having inside dimensions of 9-3/8 by 3-3/8 by 12-1/2 inches length, width, and depth, respectively. The filter shall be supported in the box by a support cell as shown on Drawing D5-3-1008. The box shall be closed in accordance with method III in the appendix to PPP-B-636.

5.3 Packing. Level shall be as specified (see 6.2).

5.3.1 Level A. The filter unit packed in accordance with 5.2.1, shall be packed, 5 per box in nailed wood boxes conforming to style 4, class 2, grade A, O to 50 lb load limit for type 1 or 2 load for overseas shipment, PPP-B-621, having inside dimensions of 18-3/8 by 13-1/4 by 9-7/8 inches length, width, and depth, respectively. As an alternate, wirebound boxes conforming to type I, class 2, grade A, MIL-B-46506 may be used. Boxes shall be assembled, closed, and strapped in accordance with the respective box specification, as applicable. Strapping used shall conform to class A or B, type I or IV, QQ-S-781.

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5.3.2 Level B. The filter, unit packed as specified, shall be packed 5 per box, in fiberboard boxes conforming to style RSC, grade W5c, of PPP-B-636, having inside dimensions of 18-3/8 by 13-1/4 by 9-7/8 inches, length, width and depth, respectively. The boxes shall be closed in accordance with Method V closure requirements and reinforced by banding with tape conforming to PPP-T-97 in accordance with the Appendix to PPP-B-636.

5.4 Unitization. When specified in the contract or order (see 6.2), filters, packed as specified in 5.3, shall. be unitized in accordance with MIL-STD-147. When quantities of packed filters are insufficient to constitute a full pallet load, they shall not be unitized.

5.5 Marking. In conjunction with information furnished in the contract, marking shall be as specified herein.

5.5.1 Unit pack marking. The outer wrap of each unit pack shall be marked in accordance with Drawing D5-3-1008.

5.5.2 Packing and unitization marking. In addition to any special markings required by the contract, shipping containers, and unitization loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This filter is intended for use with the toxicological microclimate controlled protective outfit and Filter Element Set, M13A2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit pack and packing required (see 5.2 and 5.3).
(c) The laboratory to which samples are to be sent.
(d) (Unitization, if required (see 5.4).
(e) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation before award of contract.
(2) Name and address of test facility and shipping instructions when tesing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not sample is acceptable.

6.3 Emergency breathing adapter, part no. 9391-1. Emergency Breathing Adapter, Part No. 9391-1 from Air Lock, Inc., Milford, CN has been found to be satisfactory for the leakage test (4.4.4.1).

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